



Low- and intermediate-temperature oxidation of ethylcyclohexane: A theoretical study



HongBo Ning^a, ChunMing Gong^a, NingXin Tan^a, ZeRong Li^{b,*}, XiangYuan Li^{a,*}

^a College of Chemical Engineering, Sichuan University, Chengdu 610065, PR China

^b College of Chemistry, Sichuan University, Chengdu 610064, PR China

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ABSTRACT

To get a better understanding of low temperature oxidation chemistry of long-side-chain cycloalkanes, the low- and intermediate-temperature oxidation mechanisms and kinetics of ethylcyclohexane (ECH) have been investigated by using a combination of electronic-structure theory, transition state theory (TST) and Rice-Ramsberger-Kassel-Marcus/Master-Equation (RRKM/ME) theory in this work. The high pressure limit rate constants for reactions with tight transition states are obtained by the TST, while the high pressure limit rate constants for barrierless reactions are obtained by the variational transition state theory (VTST). The rate constants in the fall-off range for pressure-dependent reactions are obtained by RRKM/ME theory. Depending on the position of the extracted H atom on the ring or on the side chain, the oxidation of ECH is mainly initiated by H-abstraction with OH radical to form cyclic C_8H_{15} radicals having six isomers (R1–R6) at low (500–900 K) and intermediate (900–1100 K) temperature and in this study we focus on the kinetics of the reactions starting from these isomers. At low temperature, it is shown that, for addition reactions of O_2 to cyclic C_8H_{15} radicals, the calculated rate constants when the unpaired electron of the radicals is in the side-chain are smaller than when in the ring. The intramolecular isomerization reactions of six cycloalkylperoxy radicals (R100–R600) by H-shift can lead to a large number of hydroperoxycycloalkyl radicals (QOOH) and it is shown that the 1,5 H-shift is more competitive than the 1,6 H-shift in the R100–R600. Furthermore, unimolecular elimination reactions of QOOH can form OH, HO_2 , cyclic ethers, ketones, aldehydes and conjugated olefins. The calculated results indicate that the formation cyclic ethers of ECH oxidation are inclined to 1-oxaspiro[3.5]octane, 8-methyl-7-oxa-bicyclo[4.2.0]octane, 7-methyl-6-oxa-bicyclo[3.2.1]octane and 2-ethyl-6-oxa-bicyclo[3.1.1]heptane. At intermediate temperature, the most important reaction type of R1–R6 consumption is the C–C bond fission to form smaller products. The reaction pathways and potential energy surfaces to form the main products (ethylene, butadiene, 1-butene, and so on) of R1–R6 decomposition are also performed and the rate constants of the corresponding reactions are calculated by using the TST. Rate constants of these reactions are fitted by a nonlinear least-squares method to the form of a modified Arrhenius rate expression, which can be directly used for the modeling study of low temperature oxidation of cycloalkanes.

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1. Introduction

There is a continued interest in developing a better understanding of the oxidation of large hydrocarbon fuels over a wide range of operating conditions. This interest is motivated by the need to improve the efficiency and performance of currently operating combustors and reduce the production of pollutant species emissions generated in the combustion process. The detailed chemical kinetic models and experiments of hydrocarbon fuels combustion are extensively

studied at the high temperature [1–2]. However, at the low temperature ($T < 900$ K), there are so many classes of elementary reactions for hydrocarbon oxidation and there are no theoretical or experimental reports on the rate constants for most of these reactions [3]. Zádor et al. [4] have reviewed the kinetics of elementary reactions in low-temperature autoignition chemistry and pointed out that the reaction kinetics of low temperature oxidation (autoignition) are still lacked in theoretical kinetics and quantum chemistry, particularly for the large reaction systems, such as the reaction of alkylcyclohexanes with O_2 . For low temperature oxidation of hydrocarbons, its heart is the chain reactions initiated by peroxy radical, ROO. It is well established that isomerization of ROO can occur via intermolecular hydrogen abstraction to form carbon-centered hydroperoxyalkyl

* Corresponding authors. Fax: +86 28 85407797.

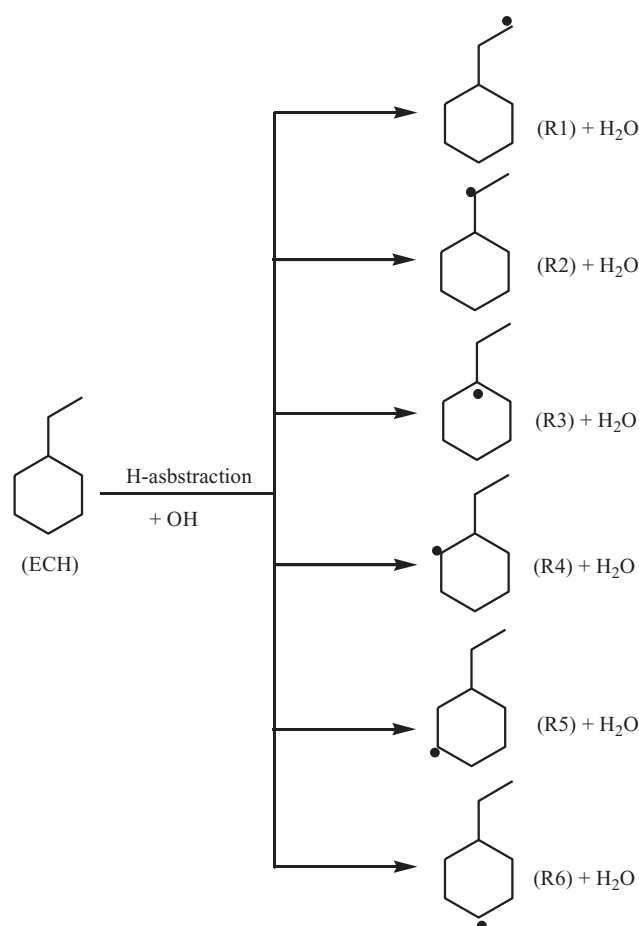
E-mail addresses: lizerong@scu.edu.cn (Z. Li), xyli@scu.edu.cn (X. Li).

radical (typically denoted as QOOH). Moreover, unimolecular decomposition of QOOH can produce a reactive OH radical and plays an important role in the radical chain propagation. Very recently, Savee et al. [5] have reported the kinetics measurements of a QOOH intermediate in the low temperature oxidation of 1,3-cycloheptadiene and experimentally confirm the existence of QOOH. Therefore, it is very necessary to study the low temperature oxidation for large hydrocarbon fuels improving efficiency for new engine designs in which ignition is controlled by fuel oxidation chemistry [6].

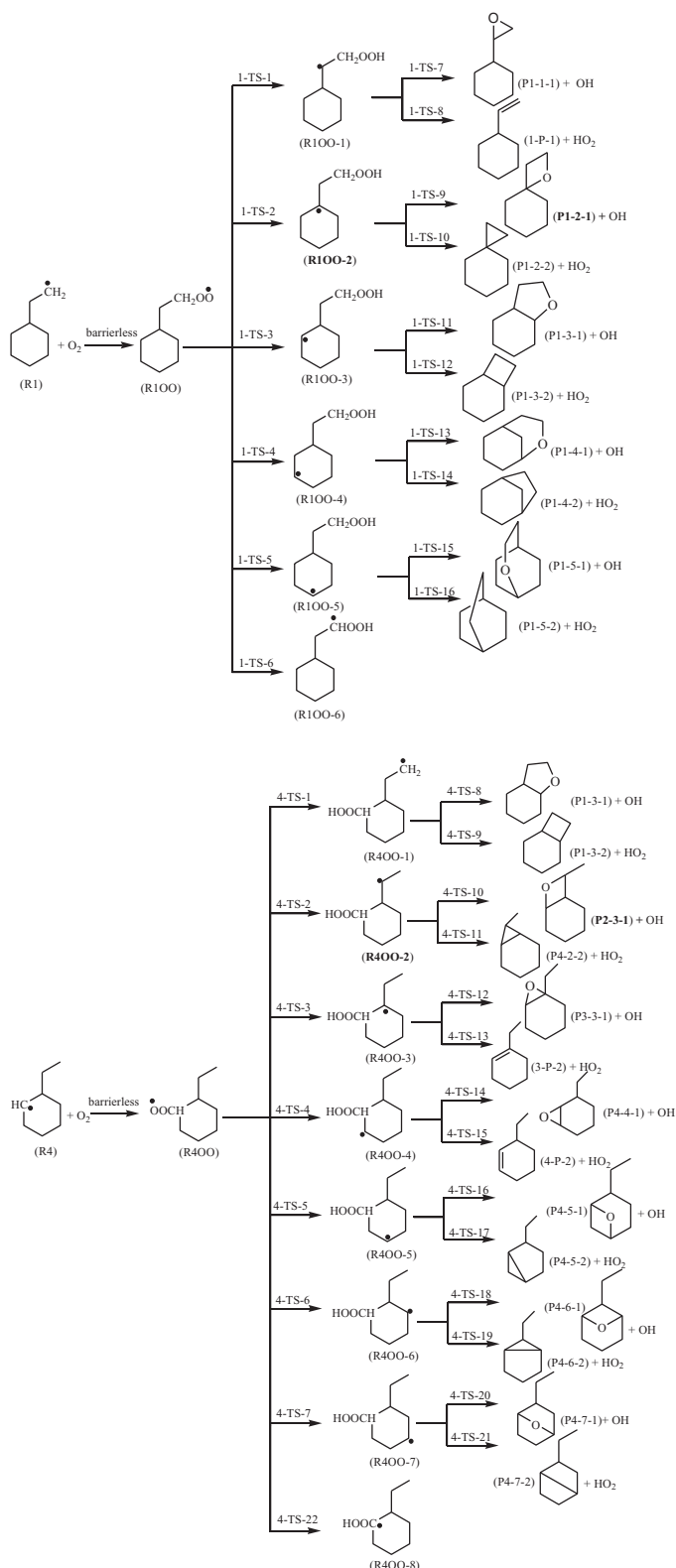
Recently, the cycloalkanes and alkylcyclohexanes have received increasing attention because of their high concentrations in practical fuels. For instance, Jet fuels such as Jet A/Jet A1/JP-8, contain up to 20 vol% of cycloalkanes, and the typical North American diesel fuels consist of up to 40 vol% of cyclohexanes and their derivatives [7–9]. Ethylcyclohexane (ECH) is a monoalkylated cyclohexane with long alkyl side-chain and also is an important model compound of surrogate fuel mixtures [10–11]. It is important to study the oxidation chemistry of these fuel components across the range of temperature and pressure conditions encountered in spark ignition and jet engines [12]. As the combustion of these compounds tends to generate a large amount of carcinogenic pollutants such as 1,3-C₄H₆, the precursors of polycyclic aromatic hydrocarbons, and toxic substances such as cyclic ethers, ketones, and aldehyde in urban environments [11–13], therefore, a better understanding of the combustion properties of the alkylcycloalkane is desired.

There have been several experimental studies on the oxidation of cyclohexane [14–16] and alkylcyclohexane such as methylcyclohexane [17–18] under different conditions. Compared to cyclohexane

and methylcyclohexane, studies on the ECH are scarce. Vanderover et al. [19] have measured the ignition delay times of a mixture of ECH/air in a shock tube ($T = 881–1319$ K; $P = 10.8–69.5$ bar; $\phi = 0.25, 0.5$ and 1). Ji et al. [20] have measured the premixed laminar flame speeds of ECH combustion in the counterflow configuration at atmospheric pressure ($T = 353$ K; $\phi = 0.7–1.5$). Later, Wu et al. [21]



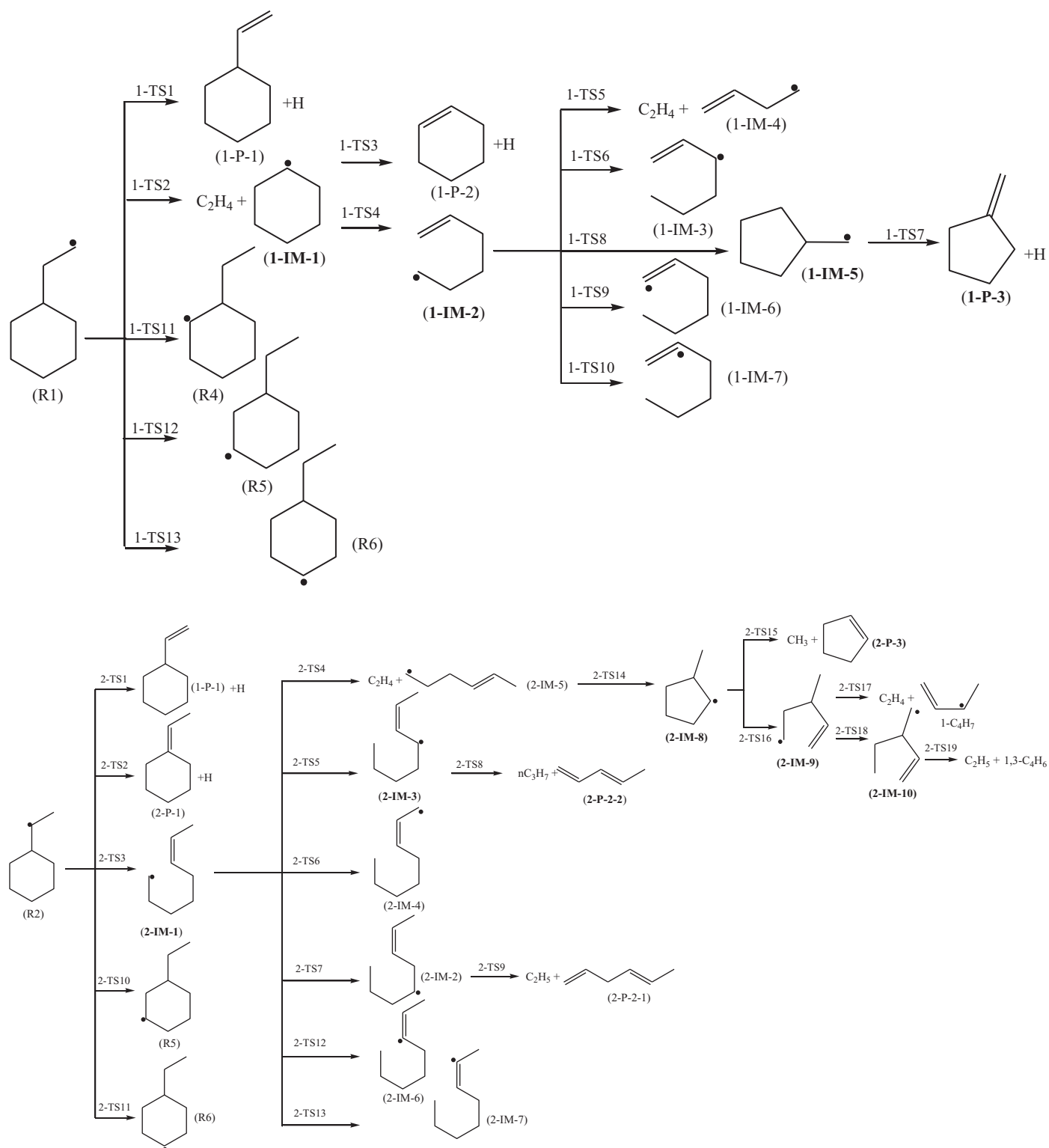
Scheme 1. The initial reactions of ECH by OH radical H-abstraction to form cyclic C₈H₁₅ radicals: R1, R2, R3, R4, R5 and R6 in the low temperature oxidation.



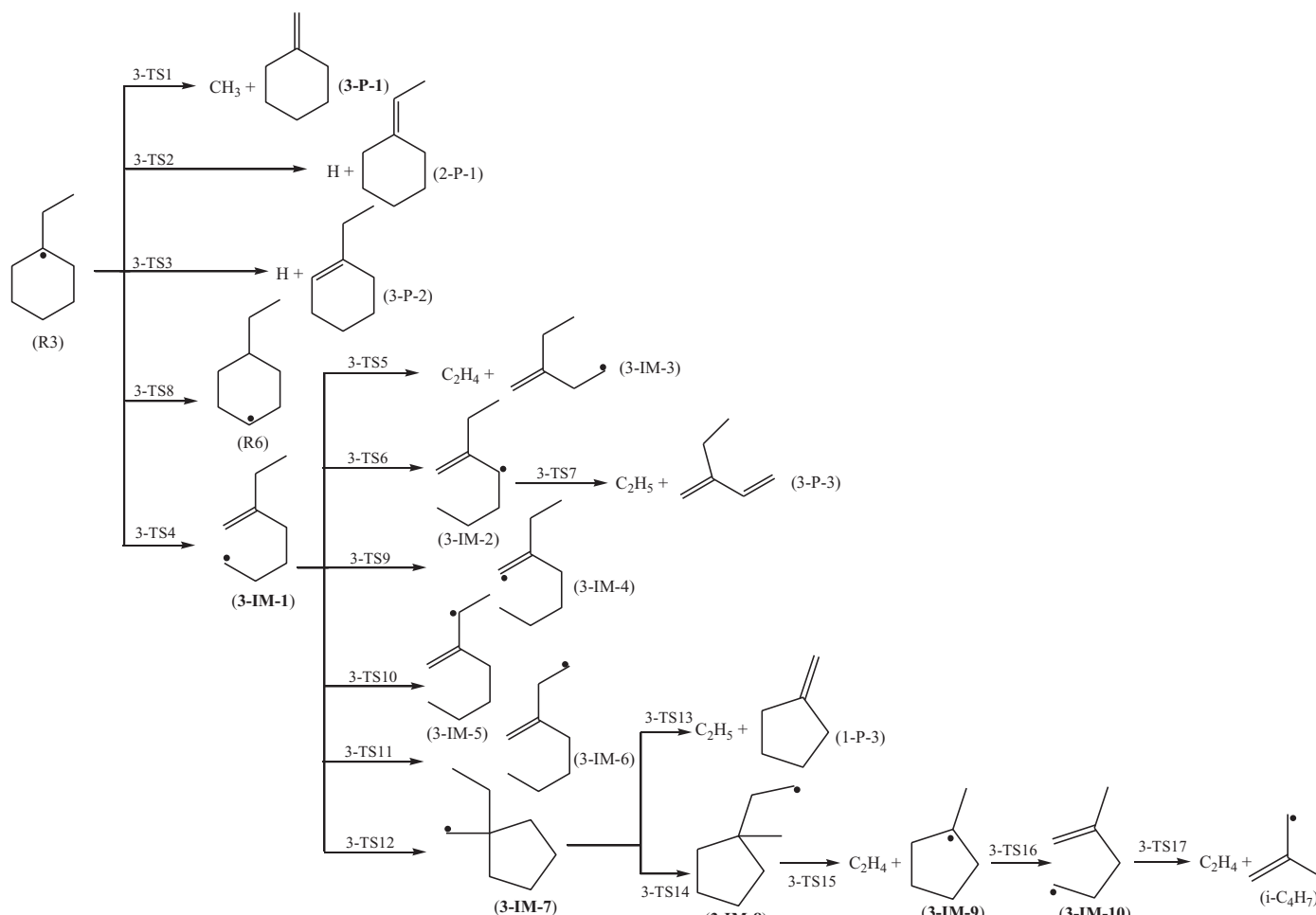
Scheme 2. The reaction pathways of R1 + O₂ and R4 + O₂ by intermolecular isomerization and β -scission reactions in the low temperature oxidation.

have measured the laminar flame speeds of cyclohexane and mono-alkylated cyclohexanes including ECH at pressures of 2 and 5 atm, and they have concluded the trend of the flame speeds: cyclohexane > n-hexane > methylcyclohexane > ethylcyclohexane at all pressures. Tian et al. [22–23] have reported the ignition delay times of ECH in a shock tube ($T = 1000$ and 1700 K; $P = 1.1$, 5.0 and 10 atm; $\phi = 0.5$, 1.0 and 2.0). An experimental study of the low- and intermediate-temperature oxidation of ECH has been performed by Husson et al.

[24] in the jet-stirred reactor ($T = 500$ – 1100 K; $P = 800$ Torr; $\phi = 0.25$, 1 and 2). In addition to 47 products having been quantified by Husson et al. [24], many species with a $C_8H_{14}O$ formula also have been detected by GC–MS in low temperature range. The main products in intermediate temperature range of ECH oxidation are also detected such as C_2H_4 , $1,3$ - C_4H_6 , 1 - C_4H_8 , and so on. Very recently, Kang et al. [25] have investigated the ignition process of ECH and its two isomers (1,2-dimethylcyclohexane and 1,3-dimethylcyclohexane) in a



Scheme 3. The reaction pathways of R1–R6 by β -scission and H-shift isomerization reactions in the intermediate temperature oxidation.



Scheme 3. Continued

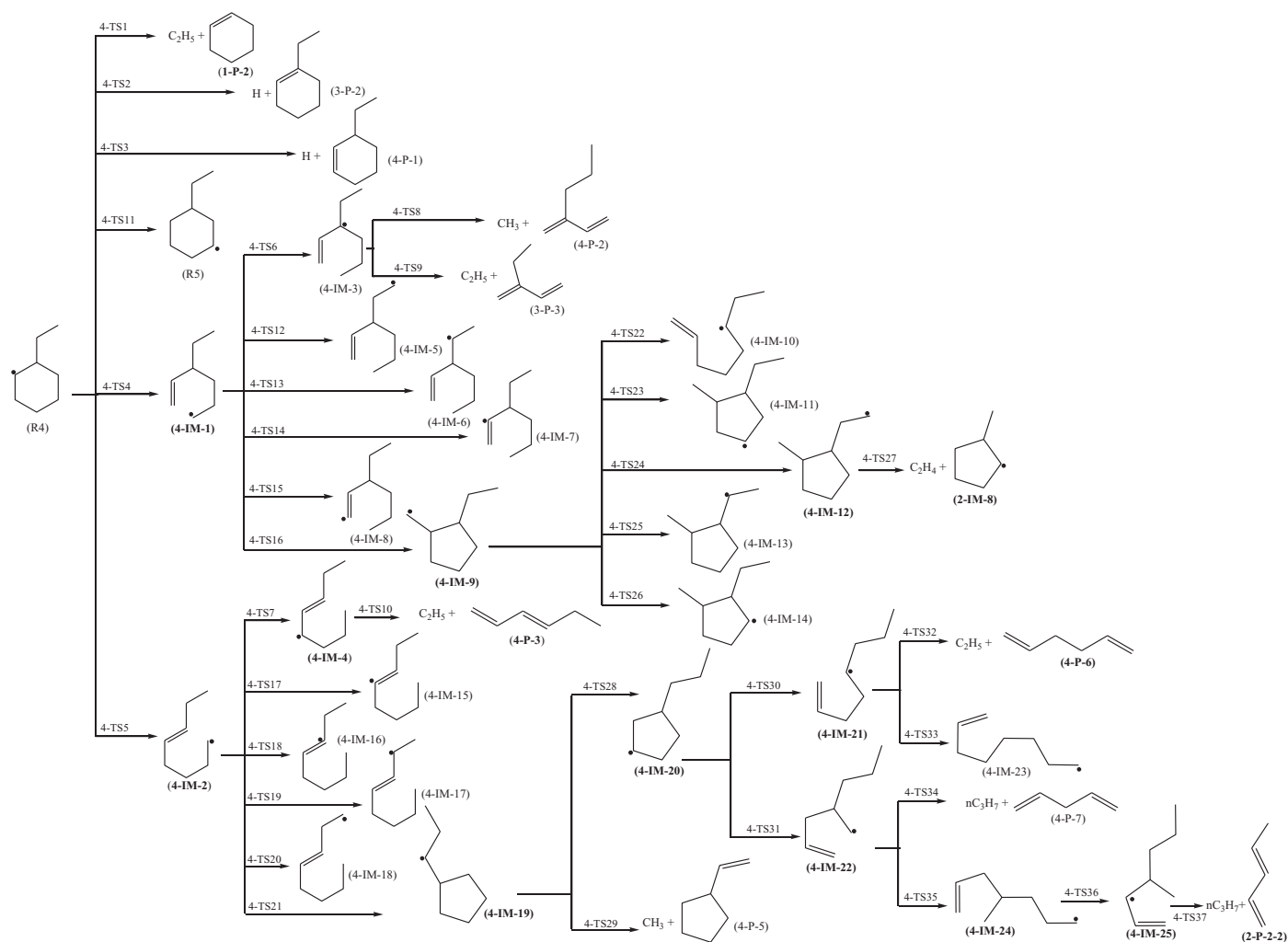
modified CFR engine ($\phi = 0.5$), and they have performed conformation analysis to explain the differences in the reactivity of the three isomers and the differences in the mole fractions of the intermediate species.

To the best of our knowledge, the study on the low temperature oxidation chemistry of n-alkylated cyclohexanes remains too weak. Very little information is available for detailed theoretical computations of the low temperature oxidation kinetics of long-side-chain cycloalkanes. The kinetic model of *JetSurF* 1.1 of n-alkane, cyclohexane, and methyl-, ethyl-, n-propyl-, n-butyl-cyclohexane oxidation has been constructed by Sirjean et al. [26] at high-temperatures. Later, the popular detailed combustion mechanism of *JetSurF* 2.0, which includes the low temperature kinetics of n-alkylated cyclohexanes oxidation, has been developed by Wang et al. [27] based on the *JetSurF* 1.1. However, the rate constants for reactions in the low temperature combustion of ECH are estimated approximately. Wang et al. [28] have investigated the primary decomposition of ECH pyrolysis using the quantum chemistry method. Recently, Wang et al. [29] have studied the experiment and detailed kinetic modeling of ECH pyrolysis and oxidation, and the kinetic model has been validated against the literature data including pyrolysis and flame, but they have not studied on the oxidation of ECH at low temperature. The objective of this work is to investigate the mechanisms and kinetics of low- and intermediate-temperature oxidation of ECH by means of quantum chemistry calculations to obtain accurate rate constants for these reactions at different temperatures and pressures which are useful in the chemical kinetic modeling of low temperature oxidation of cycloalkanes.

2. Computational details

2.1. Potential energy surface calculations

The calculations for reactants, products, intermediates, and transition states are performed by CBS-QB3 composite method [30]. In addition, many studies [31–34] have shown that the CBS-QB3 method is well adapted for the calculation of accurate energies for C/H/O system with a standard deviation of about 1.5 kcal mol⁻¹. In CBS-QB3 method, the geometry optimization as well as the vibrational analysis of molecules, is performed at the B3LYP/CBSB7 level and the single point energy calculations are performed at CCSD(T)/6-31+G(d') and MP4SDQ/CBSB4 levels. The total energy is extrapolated to the infinite-basis-set limit using pair natural-orbital energies at the MP2/CBSB3 level and an additive correction to the CCSD(T) level. In addition, CBS-QB3 includes a correction for spin contamination in open-shell species at the MP2/CBSB3 level of calculation. All the stationary points and saddle points are confirmed by frequency analysis. Moreover, intrinsic reaction coordinate (IRC) [35] calculations at the B3LYP/CBSB7 level are carried out to verify that the transition states connect the desired reactants and products. All the electronic structure calculations are performed by using Gaussian 03 program package [36]. The Cartesian coordinates of optimized structures, frequencies, rotational constants and hindered internal rotors are provided in the Supplemental Material-I. Standard enthalpies of formation ($\Delta_f H^\theta$ (298 K) kcal mol⁻¹) are calculated by using the atomization method, and the experimental values of $\Delta_f H^\theta$ (0 K) for C (169.98 kcal mol⁻¹) and H (51.63 kcal mol⁻¹) for the calculation of



Scheme 3. Continued

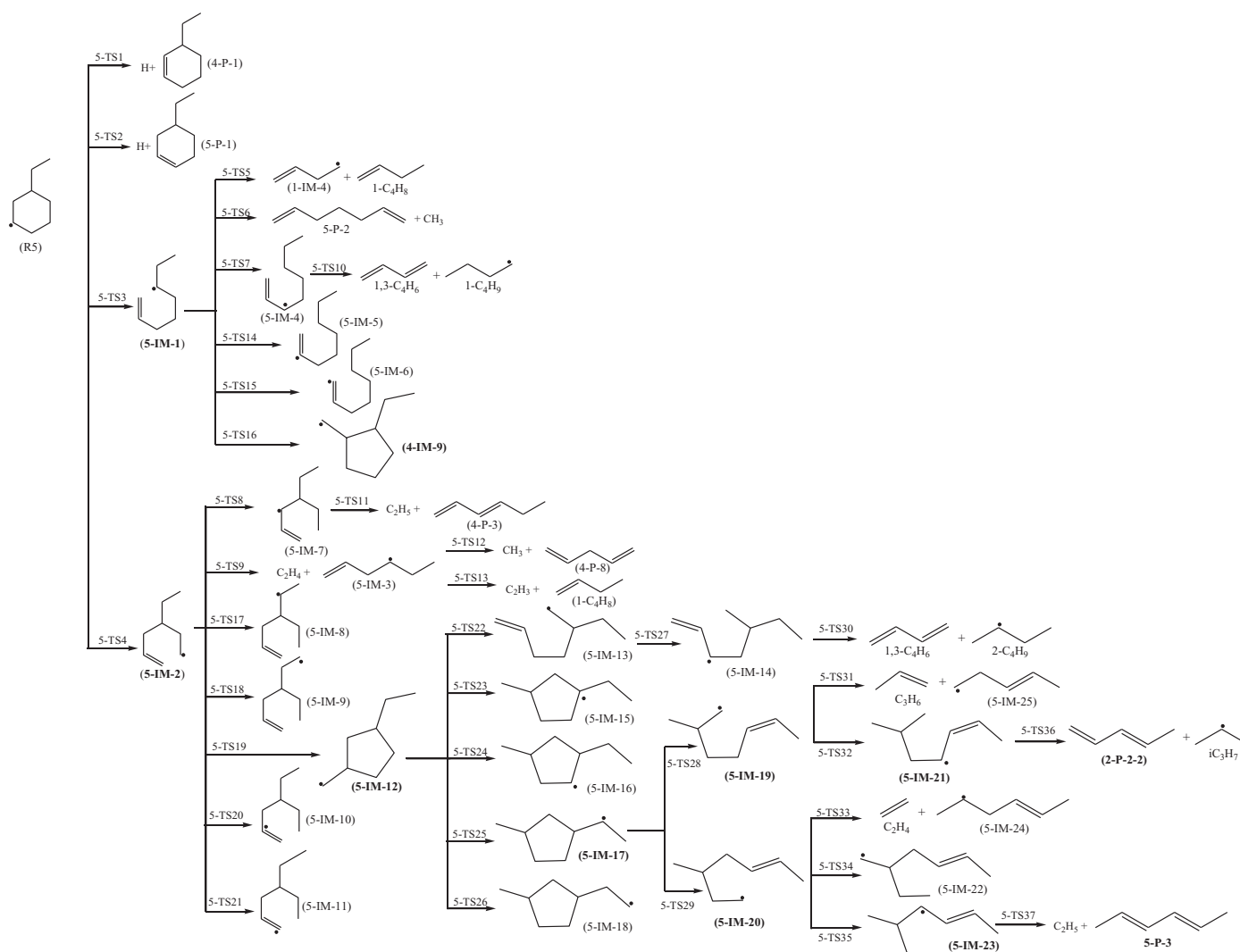
$\Delta_f H^\theta$ (0 K) are used [37]. Finally, enthalpies of formation, entropies and heat capacities of species, which have been involved in oxidation of ECH and the abbreviation symbols corresponding to the intermediate names are given in Table S1 of the Supplemental Material-II, are also listed in Table S2 of the Supplemental Material-II. Enthalpies of formation of species in this work are in close agreement with the NIST data [38].

Generally, using the rigid-rotor harmonic oscillator (RRHO) approximation to treat low-frequency vibrations can lead to significant errors in the partition function. Thus, for the intermediates and transition states, the low-frequency vibrations corresponding to torsions about a single bond are treated as one-dimensional (1-D) hindered internal rotors [39]. The potentials of each internal rotation for reactants and products are calculated at the B3LYP/CBSB7 level of theory by using a relaxed energy scan of the dihedral angle with an interval of 10° . For transition states, internal rotor scans are performed by freezing the atoms involved in the reaction coordinate and by taking into account the remaining torsions. Due to the large number of species envisaged, scan calculations are only performed for a set of reference molecules and radicals, representative of all the species considered in present work.

In this study, three reaction schemes (Schemes 1–3) are considered to describe the possible reaction pathways for low- and intermediate-temperature oxidation of ECH. Tight transition states are located at the B3LYP/CBSB7 level reactions for all β -scission

and isomerization reactions in the schemes, unless for the initial H-abstraction reactions of ECH with OH, for which we fail to locate the transition states at the B3LYP/CBSB7 level and in this particular case, the geometries optimization and frequencies calculation for the transition states are performed at the BHandHLYP/cc-pVDZ level. All single-point electronic energies are calculated by using the CBS-QB3 method.

Since the addition reaction of alkyl radical (R) and O_2 is usually barrierless, there is no well-defined saddle point. For the barrierless reactions of cyclic C_8H_{15} radicals $R1 + O_2$ and $R4 + O_2$ in Scheme 2 and $R2 + O_2$, $R3 + O_2$, $R5 + O_2$ and $R6 + O_2$ in Scheme S1 of the Supplemental Material-II. The calculations of minimum energy potential (MEP) are performed along the dissociation C–O bond at 0.1 Å intervals at the uB3LYP/CBSB7 level of theory. In each point on the MEP, the complex and reactant energy are multiplied by a scaling factor that account for the ratio of the CBS-QB3 reaction enthalpy to the uB3LYP reaction energy in order to obtain more accurate dissociation limit energy and this method has been used to study the MEP for O–H bond dissociation in benzyl hydroperoxide at the G3B3//uB3LYP/6-31G(d) level, to study the MEP for C–OO bond dissociation in the phenylperoxy radical and the *cis*- and *trans*-vinylperoxy radicals at the G3B3//O3LYP/6-31G(d) level by da Silva et al. [40–41] and to study the MEP for C–OO bond dissociation of the *tert*-isooctane hydroperoxide at the CBS-QB3//uB3LYP/6-31G(d) level by Snitsiriwat et al. [42].



Scheme 3. Continued

2.2. Rate constant calculations

All the rate constant calculations are performed by using ChemRate software [43]. For barrierless reactions the high pressure-limit rate constants are calculated with the variational transition state theory (VTST); for reactions with tight transition state, the high pressure-limit rate constants are calculated with transition state theory (TST). The time-dependent RRKM/ME theory is used to obtain pressure-dependent rate constants at pressures varying from 0.01–100 atm.

For barrierless reaction, firstly, rate constants are calculated as a function of temperature for each structure along the MEP by using the TST. Then, the rate constants obtained by the TST are minimized as a function of the reaction coordinates along the MEP to obtain the canonical variational rate constant at each temperature by the following Eq. (1):

$$k_{\text{VTST}}(T) = \min k_{\text{TST}}(s, T) \quad (1)$$

In Eq. (1), k_{TST} represents the reaction rate constant from the TST and s is the reaction coordinate. The detailed calculation procedure can be referred to the work of da Silva et al. [40–41].

For reaction with tight transition state, the high-pressure limit rate constant is calculated by using the TST method from Eq. (2):

$$k_{\infty}(T) = \kappa(T) rpd \frac{k_B T}{h} \frac{Q_{\text{TS}}(T)}{Q_{\text{R}}(T)} \exp \left[-\frac{V^{\ddagger}}{RT} \right] \quad (2)$$

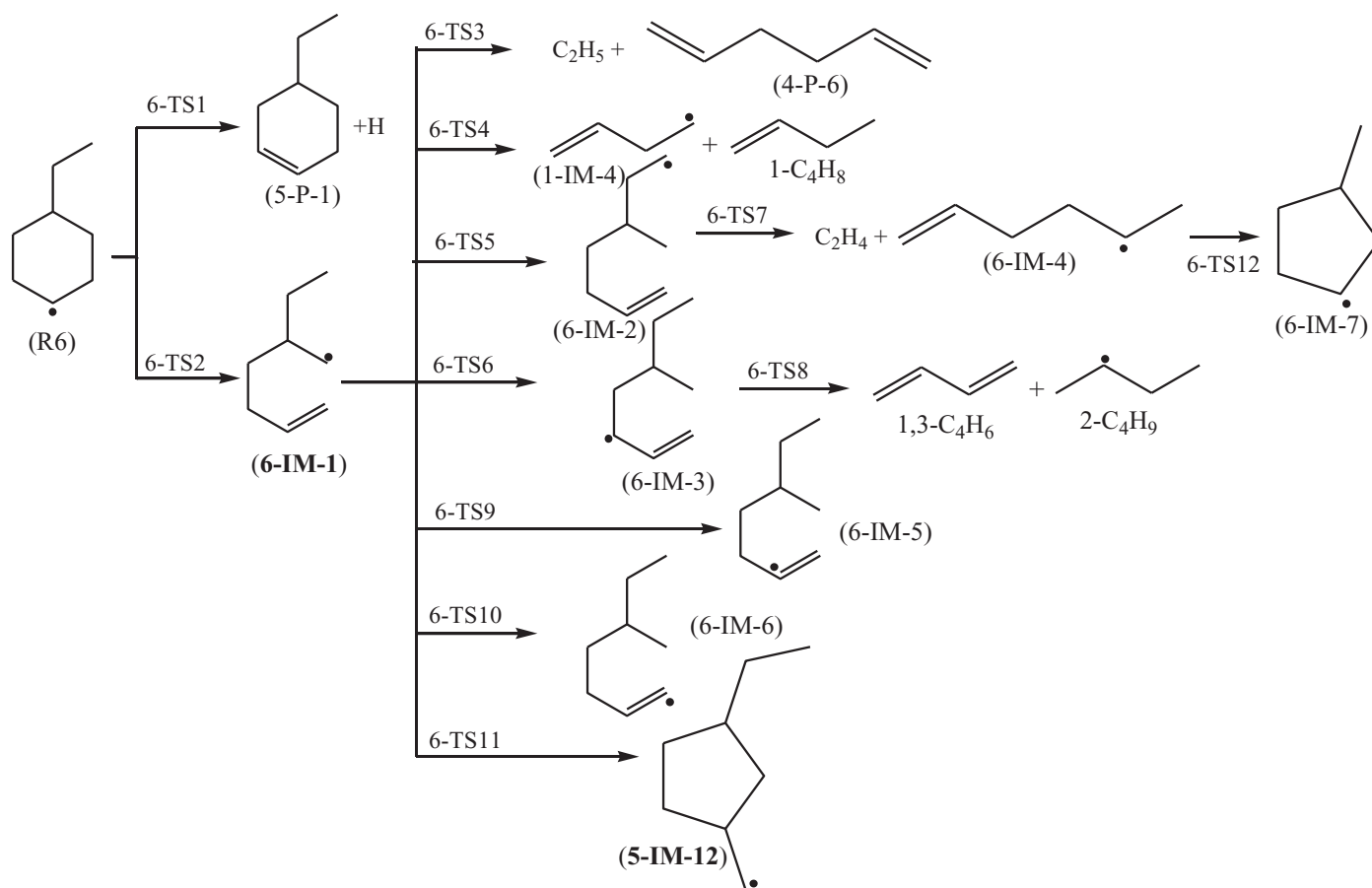
In Eq. (2), $\kappa(T)$, rpd , k_B , and h are Eckart's tunneling correction factors, the reaction path degeneracy, Boltzmann's constant, and Planck's constant, respectively. T is the temperature ranging from 300–1500 K. $Q_{\text{R}}(T)$ and $Q_{\text{TS}}(T)$ represent the partition function for the reactant (R) and the transition state (TS), respectively. $V^{\ddagger}$ is the classical barrier height including zero point energy (ZPE).

The asymmetric Eckart tunneling corrections [44] are applied to estimate the effect of 1-D tunneling through the barrier in ChemRate [43]. The barrier width L depends on three parameters, namely, E_1 , E_{-1} and $F^{\ddagger}$, which are the forward and reverse barrier heights at 0 K and force constant of the transition state, respectively. The calculated barrier width L , E_1 and E_{-1} , are given in Table S3 of the Supplemental Material-II.

The microscopic rate constants are calculated by using RRKM theory in the following Eq. (3):

$$k(E) = l^+ \left(\frac{G^+(E)}{hN(E + E_0)} \right) \quad (3)$$

In Eq. (3), $k(E)$ is the rate constant in a microcanonical ensemble with energy E . $G^+(E)$ is the total number of states of the transition state and N is the density of states of the reactant at energy $E + E_0$. l^+ and h are the reaction path degeneracy and Planck's constant,



Scheme 3. Continued

respectively. In the RRKM calculations, the density and sum of states are computed by using the Beyer–Swinehart algorithm [45]. The master-equation is used to model pressure dependent rate constants. A detailed description is presented in below about how the master equation is solved in the ChemRate program [43]. The time evolution of the population of each the species in the system is described by the master equation:

$$\frac{d\rho_i^\alpha}{dt} = \omega \sum_j P_{ij}^\alpha \rho_j^\alpha - \omega \sum_j P_{ji}^\alpha \rho_i^\alpha - k_i^\alpha \rho_i^\alpha + \sum_\beta k_l^\beta \rho_l^\beta + R_i \quad (4)$$

In Eq. (4), ρ_i^α is the molecular population of the isomer α at energy level E_i , ω is the collision frequency, which is proportional to the concentration of the third body M, hence, it is pressure dependent, P_{ij}^α is the collision energy transfer probability from energy level j to energy level i of isomer α , k_i^α is the sum of rates for all the decomposition and isomerization channels from energy level E_i , k_l^β is the rate of isomerization from isomer β at energy level E_l to isomer α at energy level E_i , and R_i is the chemical activation flux, which is assumed to be constant. Index l for isomer β corresponds to the same energy as i for α , i.e. $E_l = E_i$. The master equation can be written in the matrix form:

$$\frac{d\rho}{dt} = B\rho + R \quad (5)$$

And the overall thermal rate constant $k_{uni}(T, P)$ is equal to the negative of the largest eigenvalue of the matrix B [46]. All calculations are carried out with a series of energy grain sizes and variations in the maximum system energy to ensure that the results have converged.

For all the calculations, the energy transfer per downward collision is treated by using an exponential down model [47] with

$\langle \Delta E \rangle_{\text{down}} = 0.8 T K^{-1} \text{ cm}^{-1}$. This function form is reasonable and the similar form has been used to the decomposition of 1-pentanol [48] and reaction $\text{C}_2\text{H} + \text{C}_3\text{H}_6$ on the C_5H_7 potential energy surface [49]. The collision frequency between the reactant and bath gas (Ar) is estimated by using the Lennard-Jones (L-J) parameters σ and ϵ . The L-J parameters for species in the *JetSurF* 2.0 database [27] are taken from this database and for species not in this database are estimated on the basis of the critical temperature and pressure [50].

Finally, in order to be useful in the practical modeling using software like Chemkin-PRO [51], the modified Arrhenius form is used to describe the temperature-dependence of the rate constants and the PLOG format is used to describe the pressure-dependence (i.e., the rate parameters at any pressure P are given as a function of temperature T by functions):

$$k(T, P) = A(P)T^n(P) \exp[-E_a(P)/RT] \quad (6)$$

The rate parameters for reactions in this study are listed in the Supplemental Material- III. To facilitate this process, thermodynamic parameters for main species of ECH oxidation are also given in Chemkin format in the Supplemental Material- III.

3. Results and discussion

3.1. Potential energy surfaces

Depending on the position of the extracted H atom on the ring or on the side chain, the oxidation of ECH is mainly initiated by H-abstraction reaction with OH radical to form cyclic C_8H_{15} radicals having six isomers (R1–R6) at low (500–900 K) and intermediate (900–1100 K) temperature, which are shown in Scheme 1 and the reactions starting from these isomers are shown in Scheme 2 for

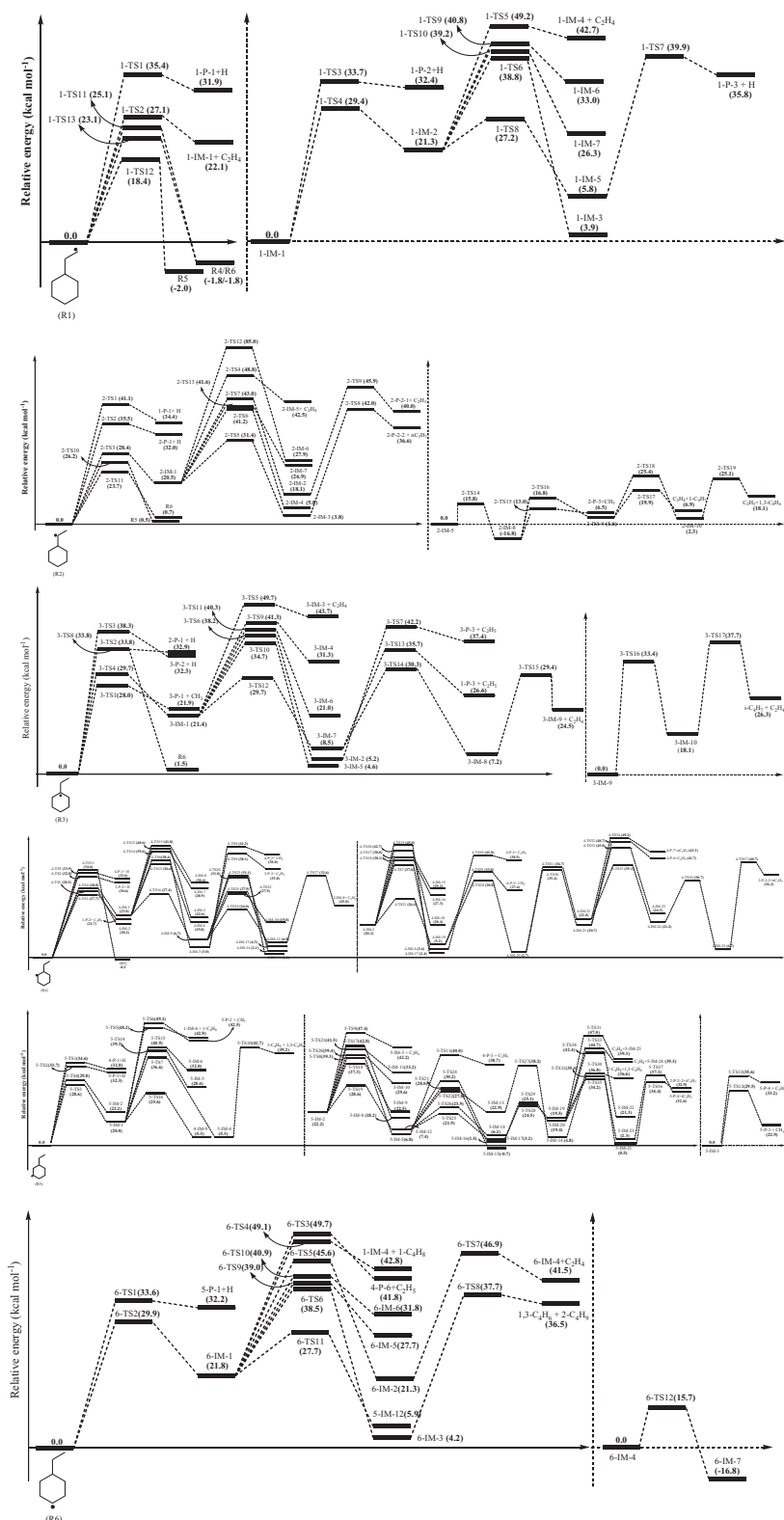


Fig. 2. Potential energy profiles for R1–R6 decomposition in the intermediate temperature oxidation at CBS-QB3 level (at 0 K including zero-point vibrational energies).

hydroperoxycycloalkyl radical (**R400-2**) by 1,5-H shift is most competitive with a barrier of 21.0 kcal mol⁻¹ and then is the OH elimination reaction to form **P2-3-1** + OH with a barrier of 17.7 kcal mol⁻¹ or the HO₂ elimination reaction to form **P4-2-2** + HO₂ with a barrier of 54.9 kcal mol⁻¹. The barriers for isomerization reactions to **R400-1**,

R400-5 and **R400-6** are 23.8, 25.1 and 24.7 kcal mol⁻¹, respectively. The most difficult step of isomerization is the isomerization reaction by 1,3-H shift to form **R400-8**, for which the barrier (38.8 kcal mol⁻¹) is much higher than that to form **R400-2** and therefore the reactions following **R400-8** are also not further studied in this study. The re-

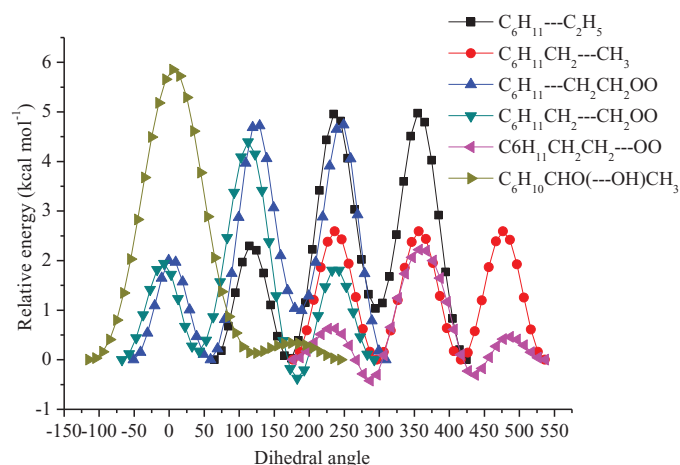


Fig. 3. Potential energy profiles for internal rotations in ECH, R1OO and R2OO-5 ($C_6H_{10}CHO(---OH)CH_3$) at B3LYP/CBSB7 level.

actions following the formation of the hydroperoxycycloalkyl radical, as shown in Scheme 2, are the decompositions of these radicals and there are two different competing channels: one is the OH elimination reaction to form O-heterocycle + OH and the other is the HO_2 elimination reaction to form di-cycloalkane + HO_2 or alkene + HO_2 . The barriers to form O-heterocycle + OH and to form alkene + HO_2 are obviously lower than HO_2 elimination to form di-cycloalkane + HO_2 . So the rate constants for the reactions of hydroperoxycycloalkyl radicals to form di-cycloalkane + HO_2 are not calculated here.

In Scheme S1 and Figure S1 of the Supplemental Material-II, the important intermediates and the products are also in bold. For R2OO, there are five possible isomers. The pathway to form hydroperoxycycloalkyl radical (**R2OO-3**) by 1,5-H shift is the lowest channel with a barrier of 21.6 kcal mol⁻¹ and then is OH elimination to produce **P2-3-1** + OH with a barrier of 23.5 kcal mol⁻¹. For R3OO, there are also five possible isomers. The pathway to form hydroperoxycycloalkyl radical (**R3OO-1**) by 1,5-H shift is the lowest channel with a barrier of 23.3 kcal mol⁻¹ and then is OH elimination to produce **P1-2-1** + OH with a barrier of 21.5 kcal mol⁻¹. For R5OO, there are seven possible isomers. The pathway to form hydroperoxycycloalkyl radical (**R5OO-6**) by 1,6-H shift is the lowest channel with a barrier of 24.5 kcal mol⁻¹ and then is OH elimination to produce **P2-4-1** + OH with a barrier of 11.9 kcal mol⁻¹. For R6OO, there are five possible

isomers. The pathway to form hydroperoxycycloalkyl radical (**R6OO-2**) by 1,5-H shift is the lowest channel with a barrier of 25.5 kcal mol⁻¹ and then is OH elimination to produce **P4-5-1** + OH with a barrier of 19.2 kcal mol⁻¹. It is found that the main cyclic ethers for oxidation of ECH are formed by 1,5-H shift of ROO and followed by OH elimination. The barrier of 1,6-H shift is smaller than 1,5-H shift in the n-alkylperoxy radicals. However, the 1,5-H shift in the cycloalkylperoxy radicals is more competitive than the 1,6-H shift.

Husson et al. [24] have experimentally reported the detailed product distributions for the low- and intermediate-temperature oxidation of ECH. In their work, many species having a $C_8H_{14}O$ formula (molecular weight of 126) are detected by GC-MS and seven of them are quantified. Among the seven species, four cyclic ethers **P1-3-1**, **P2-2-1**, **P3-3-1** and **P3-5-1** are confirmed by quantum chemistry calculations in this work. However, our calculated results indicate that the cyclic ethers **P1-2-1**, **P2-3-1**, **P2-4-1** and **P4-5-1** are more favored products than products **P1-3-1**, **P2-2-1**, **P3-3-1** and **P3-5-1** at low temperature oxidation of ECH. The pathways to form the other three species (two ketones and one aldehyde) are from the reactions between cycloalkene + OH. Zádor et al. [54] have studied the reactions between propene and hydroxyl and they found that the OH radical can add to the central carbon atom or terminal carbon atom. The main channels are the formation of vinyl alcohol + methyl group, acetone + H and acetaldehyde + CH_3 for OH adding to the central carbon atom. And the most probable products are *cis*- and *trans*-1-propenol + H, propanal + H and ethyl + CH_2O for the terminal addition of the OH radical. Analogous to the reactions between propene and hydroxyl, similar pathways for reactions between cycloalkene + OH may be possible and are shown in Scheme 4. However, the kinetics of these reactions are not studied here.

3.1.2. Intermediate temperature oxidation of ECH

The main products at intermediate temperature (900–1100 K) of ECH oxidation detected experimentally by Husson et al. [24] are ethylene, 1,3-butadiene, 1-butene, **1-P-3**, **2-P-2-2**, **3-P-1**, **3-P-3**, **4-P-3** and so on. The reaction pathways and potential energy surfaces to form the main products of ECH decomposition are performed and the rate constants of these reactions are also calculated by using TST. At intermediate temperature, the initial reactions of cyclic C_8H_{15} radicals (R1–R6) are the β -scission reactions. Ding et al. [55] have studied on a similar system by using high level of computation for the C–C fission and β -scission reactions of n-heptane. As shown in Scheme 3 and Fig. 2, R1 can form the product 1-P-1 by C–H bond fission and R1 can also form **1-IM-1** + C_2H_4 by C–C bond fission. However, the C–H bond

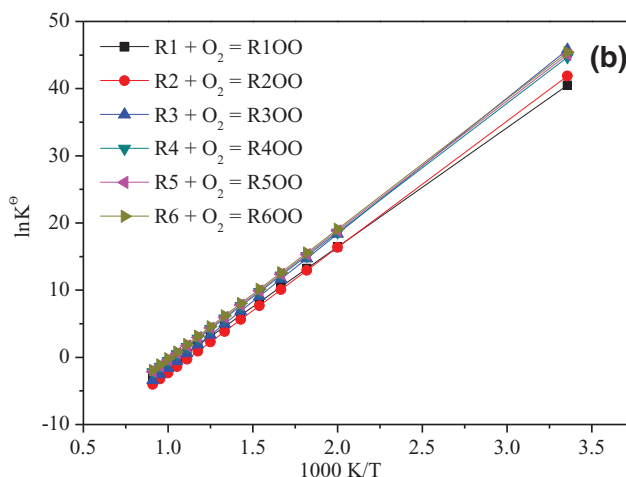
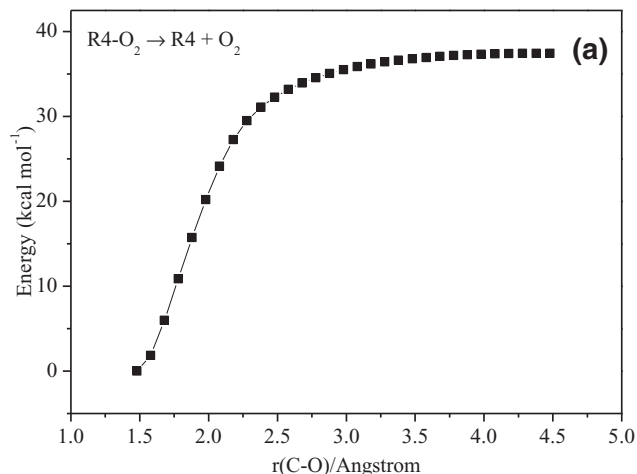
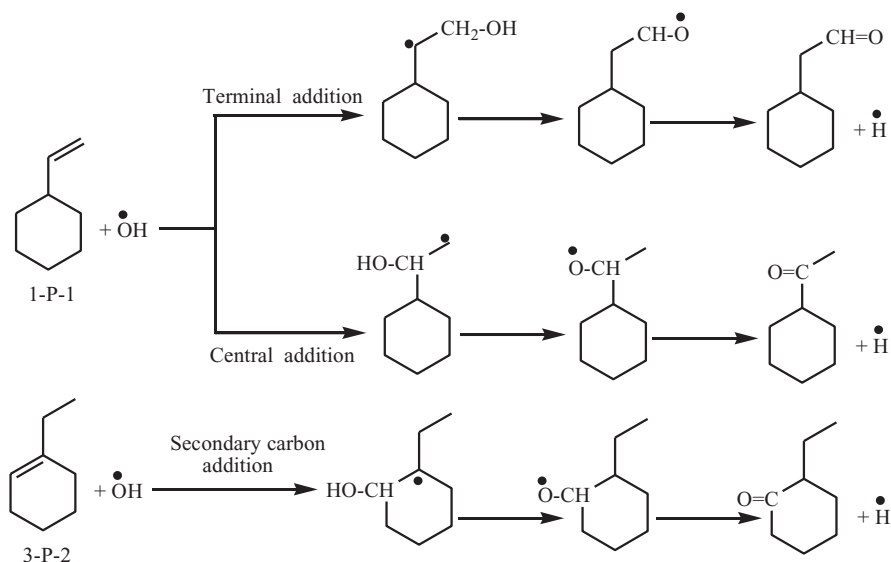


Fig. 4. (a) Potential energy profile for C–O bond dissociation of the $R4-O_2 \rightarrow R4 + O_2$ at the CBS-QB3//uB3LYP/CBSB7 level; (b) the equilibrium constants for reactions $R1 + O_2$, $R2 + O_2$, $R3 + O_2$, $R4 + O_2$, $R5 + O_2$ and $R6 + O_2$ as a function of temperature at the CBS-QB3 level.



Scheme 4. The mechanism of cycloalkene (1-P-1, 3-P-2) + OH formation to ketones and aldehyde.

fission reaction is less competitive than the C–C bond fission reaction according to the barrier height (35.4 vs 27.1 kcal mol^{−1}). In addition, isomerization of R1 can occur by 1,4-, 1,5- and 1,6-H shift reactions via a barrier height of 25.1, 18.4 and 23.1 kcal mol^{−1} to form R4, R5 and R6 radicals, respectively. Then **1-IM-1** can form **1-IM-2** by the β -scission ring-opening reaction and from **1-IM-2**, there are five competition channels: the β -scission, 5-exo ring closure and isomerization reactions by intramolecular H-shift. The channel for the 5-exo ring closure reaction of **1-IM-2** to form **1-IM-5** with a barrier of 5.9 kcal mol^{−1} is the most predominant than the channels for the β -scission reaction of **1-IM-2** to form **1-IM-4** + C₂H₄ with a barrier of 27.9 kcal mol^{−1} and the isomerization reactions of **1-IM-2** to form **1-IM-3**, **1-IM-6** and **1-IM-7** with a barrier of 17.5, 19.5 and 17.9 kcal mol^{−1}, respectively. Finally, the main product **1-P-3** is formed through the following four steps: R1 → **1-IM-1** + C₂H₄, the ring opening reaction of **1-IM-1** → **1-IM-2**, the 5-exo ring closure reaction of **1-IM-2** → **1-IM-5** and the β -scission reaction of **1-IM-5** → **1-P-3** + H.

The consumption of R2 can go through five channels and the main channels are the ring-opening reaction by the β -scission to form **2-IM-1** with a barrier of 28.4 kcal mol^{−1} and isomerization by 1,4- and 1,5-H shift reactions to form radicals R5 and R6 with a barrier of 26.2 and 23.7 kcal mol^{−1}, respectively. From **2-IM-1**, there are six competing reaction channels: the β -scission reaction to form **2-IM-5** + C₂H₄, 1,4-, 1,5-, 1,6-, 1,7- and 1,8-H shift reactions to form **2-IM-2**, **2-IM-3**, **2-IM-6**, **2-IM-7** and **2-IM-4** with a barrier of 28.3, 22.5, 10.9, 64.5, 21.1 and 20.7 kcal mol^{−1}, respectively. Obviously, the 1,5-H shift reaction of **2-IM-1** to form **2-IM-3** is the most competitive step. Thus, the main product **2-P-2-2** is formed by three steps, R2 → **2-IM-1**, **2-IM-1** → **2-IM-3** and **2-IM-3** → **2-P-2-2** + nC₃H₇. In addition, the important intermediate **2-IM-5** can occur by 5-exo ring closure to form **2-IM-8** with a barrier of 15.8 kcal mol^{−1}. From **2-IM-8**, there are two competing reaction channels: the β -scission reactions to form methyl + **2-P-3**, **2-IM-9** with a barrier height of 29.8 and 33.6 kcal mol^{−1}, respectively. Then **2-IM-9** can form C₂H₄ + 1-C₄H₇ by β -scission reaction and can also form **2-IM-10** by 1,4-H shift with a barrier of 18.3 and 23.8 kcal mol^{−1}, respectively. Finally, the **2-IM-10** can form the main product 1,3-C₄H₆ by β -scission reaction with a barrier of 22.8 kcal mol^{−1}.

The decomposition of R3 can go through five channels and the CH₃ elimination reaction to form **3-P-1** + CH₃ and the ring-opening reaction to form **3-IM-1** are the main reaction channels with a barrier of 28.0 and 29.7 kcal mol^{−1}, respectively. The further reactions

of **3-IM-1** can form **3-IM-2**, **3-IM-4**, **3-IM-5** and **3-IM-6** by 1,4-, 1,6-, 1,6- and 1,7-H shift reactions with a barrier of 16.8, 19.9, 13.3 and 18.9 kcal mol^{−1}, respectively, and can also form **3-IM-3** + C₂H₄ by β -scission with a barrier of 28.3 kcal mol^{−1} and form **3-IM-7** by 5-exo ring closure with a barrier of 8.3 kcal mol^{−1}. Obviously, the 5-exo ring closure of **3-IM-1** to form **3-IM-7** is the most predominant step. From **3-IM-7**, it can form C₂H₅ + 1-P-3, **3-IM-8** by β -scission and 1,4-H shift reactions with a barrier of 27.2 and 21.8 kcal mol^{−1}, respectively. Then **3-IM-8** can form C₂H₄ + **3-IM-9** with a barrier of 22.2 kcal mol^{−1}. Finally, the main product C₂H₄ can be formed by two steps: **3-IM-9** → **3-IM-10** and **3-IM-10** → C₂H₄ + i-C₄H₇.

R4 can be consumed through six channels. The calculated results indicate that the main channels for R4 consumption are C₂H₅ elimination reaction to form **1-P-2** + C₂H₅, the ring-opening reactions to form **4-IM-1** and **4-IM-2** with a barrier of 27.7, 28.8 and 28.5 kcal mol^{−1}, respectively. The further reactions of **4-IM-1** can occur by 1,4-, 1,5-, 1,5-, 1,6-, 1,6-H shift and 5-exo ring closure to form **4-IM-3**, **4-IM-6**, **4-IM-7**, **4-IM-5**, **4-IM-8** and **4-IM-9** with a barrier of 15.4, 13.6, 16.6, 17.6, 18.8 and 4.4 kcal mol^{−1}, respectively. Obviously, the 5-exo ring closure of **4-IM-1** to form **4-IM-9** is the most important step. The further reactions of **4-IM-9** can form **4-IM-10** by β -scission reaction with a barrier of 20.8 kcal mol^{−1} and form **4-IM-11**, **4-IM-14**, **4-IM-13** and **4-IM-12** by 1,4-, 1,4-, 1,4- and 1,5-H shift reactions with a barrier of 24.4, 24.3, 21.2 and 17.3 kcal mol^{−1}, respectively. Then the **4-IM-12** can form C₂H₄ + **2-IM-8** by β -scission reaction with a barrier of 25.7 kcal mol^{−1}. The detailed reaction channels of **2-IM-8** have been discussed in above R2 radical section. In addition, the further reactions of the other important radical **4-IM-2** can occur by similar 1,4-, 1,5-, 1,6-, 1,7-, 1,8-H shift and 5-exo ring closure to form **4-IM-4**, **4-IM-15**, **4-IM-16**, **4-IM-17**, **4-IM-18** and **4-IM-19** with a barrier of 16.8, 18.6, 17.9, 24.8, 22.5 and 6.2 kcal mol^{−1}, respectively. Obviously, the 5-exo ring closure reaction of **4-IM-2** to form **4-IM-19** is the most important channel. Then the **4-IM-19** can form CH₃ + **4-P-5** by β -scission reaction with a barrier of 28.6 kcal mol^{−1} and can also form **4-IM-20** by 1,4-H shift reaction with a barrier of 25.6 kcal mol^{−1}. Finally, the main product **2-P-2-2** can be formed by four steps: **4-IM-20** → **4-IM-22**, **4-IM-22** ↔ **4-IM-24**, **4-IM-24** ↔ **4-IM-25** and **4-IM-25** → nC₃H₇ + **2-P-2-2** with a barrier of 34.0, 17.1, 9.5 and 36.0 kcal mol^{−1}, respectively.

There are four channels for decomposition of R5. The two β -scission ring-opening reactions of R5 to form **5-IM-1** and **5-IM-2** with a barrier of 28.6 and 29.8 kcal mol^{−1}, respectively, are more

predominant than the H elimination reaction steps to form 5-P-1 + H and 4-P-1 + H with a barrier of 33.7 and 34.6 kcal mol⁻¹, respectively. The first ring-opening reaction produces **5-IM-1**, which undergoes 1,4-, 1,5-, 1,6-H shift reactions to form 5-IM-4, 5-IM-5, 5-IM-6 and undergoes β -scission reactions to form 1-IM-4 + 1-C₄H₈ and 5-P-2 + CH₃ and also occur by 5-exo ring closure to form **4-IM-9** with a barrier of 16.6, 19.3, 20.9, 28.2, 29.1 and 5.6 kcal mol⁻¹, respectively. Obviously, the 5-exo ring closure of **5-IM-1** to form **4-IM-9** is the most important channel and the detailed reaction channels of **4-IM-9** have been discussed in above R4 radical section. The other important intermediate **5-IM-2** can occur by 1,4-, 1,4-, 1,5-, 1,5-, 1,6-H shift, β -scission and 5-exo ring closure reactions to form 5-IM-7, 5-IM-8, 5-IM-9, 5-IM-10, 5-IM-11, C₂H₄ + 5-IM-3 and **5-IM-12** with a barrier of 17.1, 20.6, 15.1, 17.2, 19.3, 25.2 and 6.4 kcal mol⁻¹, respectively. The 5-exo ring closure of **5-IM-2** to form **5-IM-12** is the most important channel. From **5-IM-12**, it can form 5-IM-13, 5-IM-15, 5-IM-16, **5-IM-17** and 5-IM-18 by β -scission, 1,4-, 1,4-, 1,5- and 1,6-H shift reactions with a barrier of 20.4, 20.6, 22.8, 14.5 and 16.5 kcal mol⁻¹, respectively. Then the **5-IM-17** can occur by β -scission reactions to form **5-IM-19** and **5-IM-20** with a barrier of 21.3 and 21.9 kcal mol⁻¹, respectively. Finally, the main products **2-P-2-2** and **5-P-3** can be formed by two steps: **5-IM-19** $\leftrightarrow$ **5-IM-21**, **5-IM-21** $\rightarrow$ **2-P-2-2** + iC₃H₇; **5-IM-20** $\leftrightarrow$ **5-IM-23**, **5-IM-23** $\rightarrow$ **5-P-3** + C₂H₅, with a barrier of 15.9, 33.9 kcal mol⁻¹; 14.8, 34.8 kcal mol⁻¹, respectively.

There are two channels for R6 consumption. The calculated energy barriers indicate that the main channel from R6 is the formation of **6-IM-1** by the β -scission ring-opening reaction with a barrier of 29.9 kcal mol⁻¹. **6-IM-1** can go through seven competing reaction channels by the β -scission, 1,4-, 1,5-, 1,6-H shift and 5-exo ring closure reactions: **6-IM-1** $\rightarrow$ 4-P-6 + C₂H₅, **6-IM-1** $\rightarrow$ 1-IM-4 + 1-C₄H₈, **6-IM-1** $\leftrightarrow$ 6-IM-2, **6-IM-1** $\leftrightarrow$ 6-IM-3, **6-IM-1** $\leftrightarrow$ 6-IM-5, **6-IM-1** $\leftrightarrow$ 6-IM-6 and **6-IM-1** $\rightarrow$ **5-IM-12** with a barrier of 27.9, 27.3, 23.8, 16.7, 17.2, 19.1 and 5.9 kcal mol⁻¹, respectively. The most competitive channel is the formation of **5-IM-12** by 5-exo ring closure reaction from **6-IM-1**. The detailed reaction channels of intermediate **5-IM-12** have been discussed in above R5 radical section.

For the decomposition of R1, R2, R3, R4, R5 and R6 radicals, it can be seen that the 5-exo ring closure is the most important reaction channel and the main products of ECH oxidation at intermediate temperature are mainly formed by the isomerization and β -scission reactions of the intermediates which are formed by 5-exo ring closure reaction.

3.2. Rate constants

3.2.1. High-pressure limit rate constants

In this study, the rate constants are calculated only for reactions with lower barrier heights when there are many competing reaction channels and there are 92 reactions in this study. The high-pressure limit rate constants for these reactions are calculated by using VTST and TST methods and the results are compared with the kinetic model of *JetSurF 2.0* [27]. The rate constants at the high-pressure limit in the temperature range of 300–1500 K are given in Table 1. The rate constant uncertainties are determined by the accuracies of the energy barriers, vibrational frequencies, collisional energy transfer parameters, hindered rotor barrier heights and the Lennard-Jones (L-J) parameters, etc. Hai Wang et al. [56] have detailed reviewed the rate constant uncertainties and the combustion kinetic model uncertainty quantification, propagation and minimization and also emphasize that the kinetic uncertainty must be considered and treated as a part of the combustion chemistry development in order to make progress. Very recently, Xing et al. [57] have studied the global uncertainty analysis for RRKM/master equation based kinetic predictions for a case of ethanol decomposition and have concluded the high dimensional model representation (HDRM) which provides a great tool for investigating the propagation of uncertainty through RRKM/ME

Table 1

High-pressure limit rate parameters in low temperature oxidation of ECH in the temperature range 300–1500 K.

Reactions	Log A	n	E _a (cal mol ⁻¹)
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R1 + H ₂ O	-20.34	3.12	4127.9
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R2 + H ₂ O	-20.23	3.11	1856.6
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R3 + H ₂ O	-19.37	2.88	447.4
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R4 + H ₂ O	-19.95	3.13	1352.4
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R5 + H ₂ O	-20.04	3.11	2015.2
cy-C ₆ H ₁₁ C ₂ H ₅ + OH $\rightarrow$ R6 + H ₂ O	-20.01	3.11	2118.4
R100 $\rightarrow$ R1 + O ₂	11.09	0.53	31862.8
R200 $\rightarrow$ R2 + O ₂	19.27	-1.70	34438.1
R300 $\rightarrow$ R3 + O ₂	22.41	-2.19	39017.8
R400 $\rightarrow$ R4 + O ₂	16.67	-0.75	33812.4
R500 $\rightarrow$ R5 + O ₂	24.54	-3.07	37162.9
R600 $\rightarrow$ R6 + O ₂	23.67	-2.85	37309.0
R100 $\leftrightarrow$ R100-2	4.86	2.39	12631.4
R200 $\leftrightarrow$ R200-3	4.08	2.66	16661.4
R200 $\leftrightarrow$ R200-4	1.50	-4.61	17975.6
R200 $\leftrightarrow$ R200-5	2.32	3.14	19199.4
R300 $\leftrightarrow$ R300-1	4.99	2.50	18779.1
R300 $\leftrightarrow$ R300-4	0.36	3.72	18950.9
R400 $\leftrightarrow$ R400-1	3.33	2.69	18617.7
R400 $\leftrightarrow$ R400-2	4.48	2.53	16311.9
R400 $\leftrightarrow$ R400-5	0.70	3.61	18434.1
R400 $\leftrightarrow$ R400-6	0.47	3.65	18100.3
R500 $\leftrightarrow$ R500-3	-4.70	5.19	17914.3
R500 $\leftrightarrow$ R500-4	-0.24	3.83	18512.8
R500 $\leftrightarrow$ R500-6	0.20	3.66	17722.8
R500 $\leftrightarrow$ R500-7	0.13	3.45	21956.6
R600 $\leftrightarrow$ R600-2	-0.16	3.72	18608.9
R600 $\leftrightarrow$ R600-3	-4.69	5.15	18942.8
R100-2 $\rightarrow$ P1-2-1 + OH	-1.99	4.24	3523.8
R200-3 $\rightarrow$ P2-3-1 + OH	-12.43	7.08	9301.8
R300-1 $\rightarrow$ P1-2-1 + OH	-20.50	9.56	1125.1
R400-2 $\rightarrow$ P2-3-1 + OH	-9.70	6.61	3778.7
R500-6 $\rightarrow$ P2-4-1 + OH	1.05	3.32	4873.0
R600-2 $\rightarrow$ P4-5-1 + OH	-12.39	7.29	3921.7
R1 $\leftrightarrow$ R4	-11.07	6.68	11888.3
R1 $\leftrightarrow$ R5	-1.37	3.99	10167.5
R1 $\leftrightarrow$ R6	-2.61	4.40	14321.8
R2 $\leftrightarrow$ R5	-13.11	7.40	12489.0
R2 $\leftrightarrow$ R6	-6.90	5.63	13360.2
R1 $\rightarrow$ 1-IM-1 + C ₂ H ₄	9.90	1.05	25956.6
1-IM-1 $\rightarrow$ 1-IM-2	7.93	1.74	27621.5
1-IM-2 $\rightarrow$ 1-IM-5	6.30	1.15	2562.5
1-IM-5 $\rightarrow$ 1-P-3 + H	7.85	1.30	32794.4
R2 $\rightarrow$ 2-IM-1	7.92	1.53	26612.6
2-IM-1 $\leftrightarrow$ 2-IM-3	2.95	2.42	5101.1
2-IM-3 $\rightarrow$ 2-P-2-2 + nC ₃ H ₇	10.39	0.85	38310.1
2-IM-8 $\rightarrow$ CH ₃ + 2-P-3	8.32	1.81	28133.4
2-IM-8 $\rightarrow$ 2-IM-9	6.41	2.10	30741.0
2-IM-9 $\leftrightarrow$ 2-IM-10	-6.89	5.49	12631.6
2-IM-10 $\rightarrow$ C ₂ H ₅ + 1,3-C ₄ H ₆	7.99	1.76	20514.3
R3 $\rightarrow$ 3-P-1 + CH ₃	7.07	2.04	26178.0
R3 $\rightarrow$ 3-IM-1	7.95	1.73	27960.2
3-IM-1 $\leftrightarrow$ 3-IM-2	-6.09	4.98	7331.8
3-IM-2 $\rightarrow$ 3-P-3 + C ₂ H ₅	12.08	0.55	37010.3
3-IM-1 $\rightarrow$ 3-IM-7	6.20	1.30	4598.9
3-IM-7 $\leftrightarrow$ 3-IM-8	-6.38	5.27	10958.8
3-IM-8 $\rightarrow$ 3-IM-9 + C ₂ H ₄	9.58	1.14	20564.0
3-IM-9 $\rightarrow$ 3-IM-10	5.67	2.27	30400.2
3-IM-10 $\rightarrow$ i-C ₄ H ₇ + C ₂ H ₄	1.85	3.44	13382.3
R4 $\rightarrow$ 1-P-2 + C ₂ H ₅	9.36	1.32	26383.2
R4 $\rightarrow$ 4-IM-2	8.24	1.84	26688.3
4-IM-2 $\leftrightarrow$ 4-IM-4	-3.51	4.42	7397.3
4-IM-4 $\rightarrow$ 4-P-3 + C ₂ H ₅	10.20	1.21	37542.7
4-IM-1 $\rightarrow$ 4-IM-9	8.38	0.72	1771.4
4-IM-9 $\leftrightarrow$ 4-IM-12	-0.14	3.37	10004.9
4-IM-12 $\rightarrow$ C ₂ H ₄ + 2-IM-8	8.55	1.54	23563.3
4-IM-2 $\rightarrow$ 4-IM-19	7.77	0.95	2984.8
4-IM-19 $\leftrightarrow$ 4-IM-20	-12.97	7.00	11689.4
4-IM-20 $\rightarrow$ 4-IM-21	7.69	1.76	30321.7
4-IM-20 $\rightarrow$ 4-IM-22	6.61	2.02	31343.3
4-IM-22 $\leftrightarrow$ 4-IM-24	-0.72	3.47	9320.8
4-IM-24 $\leftrightarrow$ 4-IM-25	3.33	2.26	4169.8
4-IM-25 $\rightarrow$ nC ₃ H ₇ + 2-P-2-2	10.78	1.31	35579.1
R5 $\rightarrow$ 5-IM-1	8.98	1.42	27117.2

(continued on next page)

Table 1 (continued)

Reactions	Log A	n	E_a (cal mol ⁻¹)
R5 → 5-IM-2	8.81	1.91	28670.8
5-IM-1 ↔ 5-IM-4	-7.25	5.49	6994.2
5-IM-2 ↔ 5-IM-7	-5.24	4.82	7853.3
5-IM-4 → 1-C ₄ H ₉ + 1,3-C ₄ H ₆	7.06	1.67	36000.5
5-IM-7 → 4-P-3 + C ₂ H ₅	9.60	1.13	33155.6
5-IM-1 → 4-IM-9	7.18	0.88	2689.7
5-IM-2 → 5-IM-12	8.36	0.90	3545.4
5-IM-12 ↔ 5-IM-17	-1.05	3.65	8058.5
5-IM-17 → 5-IM-19	7.79	1.49	19214.5
5-IM-17 → 5-IM-20	7.53	1.60	19680.6
5-IM-19 ↔ 5-IM-21	-2.64	4.18	7104.8
5-IM-20 ↔ 5-IM-23	-1.78	3.97	6431.0
5-IM-21 → iC ₃ H ₇ + 2-P-2-2	11.21	0.99	34,036.6
5-IM-23 → C ₂ H ₅ + 5-P-3	10.30	1.67	34,323.9
R6 → 6-IM-1	8.35	1.65	28306.6
6-IM-1 ↔ 6-IM-3	-3.29	4.33	7402.4
6-IM-3 → 2-C ₄ H ₉ + 1,3-C ₄ H ₆	9.94	1.11	33587.9
6-IM-1 → 5-IM-12	8.06	0.86	2903.6

calculations. Here, we only consider the influence of the reaction energy barriers for the rate constant uncertainty. The couple cluster CCSD(T) method target a chemical accuracy of ± 1 kcal mol⁻¹ for reaction energy barriers [58–59], which accuracy gives an expected uncertainty factor value of 1.65, 1.4 and 1.3 in the rate constant at 1000 K, 1500 K and 2000 K, respectively. The CBS-QB3 method target the chemical accuracy of ± 1.5 kcal mol⁻¹ for reaction energy barriers, which accuracy gives an expected uncertainty factor value of

2.12, 1.65 and 1.5 in the rate constant at 1000 K, 1500 K and 2000 K, respectively.

In the literature, there are few studies on the rate constants for the association reaction of cycloalkyl R + O₂. Knepp et al. [60] have reported the rate constant value of 9.94×10^{12} cm³ mol⁻¹ s⁻¹ at 298 K for the association reaction of cyclohexyl with O₂, which is calculated by using the VRC-TST theory. The rate constant values of reactions R1 + O₂, R2 + O₂, R3 + O₂, R4 + O₂, R5 + O₂ and R6 + O₂ at 300 K in this study are 2.34×10^{11} , 3.65×10^{12} , 3.52×10^{13} , 5.63×10^{13} , 8.04×10^{13} and 3.58×10^{13} cm³ mol⁻¹ s⁻¹, respectively. Figure 5(a) shows a comparison of the high-pressure limit rate constants for these reactions calculated in this work with the kinetic model of *JetSurF* 2.0 [27] and the work of Knepp et al. In the *JetSurF* 2.0, the rate constants for these six reactions are estimated by an approximate method and a same value is assigned for them. Our calculated results show that only the rate constant of reaction R3 + O₂ agrees well with that from *JetSurF* 2.0 with a factor of 1.5–2.5. The rate constants of the other reactions are all lower than the values of *JetSurF* 2.0 above 500 K, moreover the rate constants of R1 + O₂ and R2 + O₂ reactions are lower by one to two orders of magnitude than the values of *JetSurF* 2.0, which may explain why the consumption of the ECH modeled by *JetSurF* 2.0 is faster the experimental results in the Husson et al. [24] work and hence our calculated results can be helpful to improve the *JetSurF* 2.0. In addition, we also find that the calculated rate constants for addition reactions R1 + O₂ and R2 + O₂ where the unpaired electron of cyclic C₈H₁₅ radicals is in the side-chain are smaller than those for R3 + O₂, R4 + O₂, R5 + O₂ and R6 + O₂ which the unpaired electron of cyclic C₈H₁₅ radicals is in the ring. For the 5-exo ring closure reaction 1-IM-2 → 1-IM-5, Fig. 5(b) shows a comparison of the

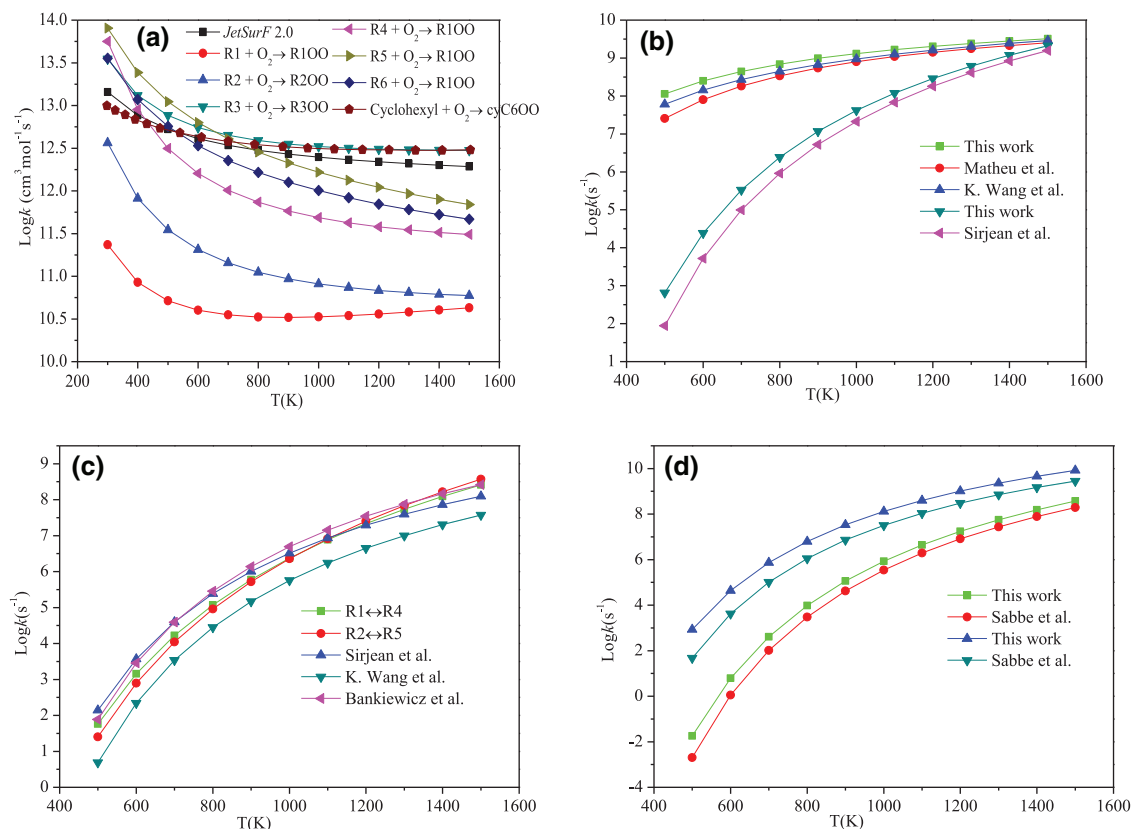


Fig. 5. Plots of the calculated high pressure limit rate constants: (a) addition reactions for O₂ adding to the cyclic C₈H₁₅ radicals (R1–R6), the estimated rate constant of *JetSurF* 2.0 [27] and the calculated rate constant for reaction cyclohexyl + O₂ = cycC₆OO by Knepp et al. [60]; (b) 5-exo ring closure reaction 1-IM-2 → 1-IM-5 in this work, Matheu et al. [61] and Wang et al. [63] and 1-IM-5 → 1-IM-2 in this work and Sirjean et al. [62]; (c) 1,4-H shift reactions R1 ↔ R4 and R2 ↔ R5 in this work, C-CCC ↔ CCCC (Wang et al. [63]), 1-heptyl ↔ 4-heptyl (Sirjean et al. [64]) and C-CCC ↔ CCCC (Bankiewicz et al. [65]); (d) β-scission reactions for 2-IM-11 → C₂H₅ + 1,3-C₄H₆ (in blue) and 6-IM-3 → 2-C₄H₉ + 1,3-C₄H₆ (in green) and CH₂ = CHC(CH₃)CH₂CH₂ → CH₃ + CH₂ = C(CH₃)CH = CH₂ (in cyan) and CH₂ = CHCH(CH₃)CH₂ → CH₃ + 1,3-C₄H₆ (in red) by Sabbe et al. [66]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

calculated high-pressure limit rate constants in this work with the values by Matheu et al. [61], Sirjean et al. [62] and K. Wang et al. [63]. It is shown that our calculated rate constants for 1-IM-2→1-IM-5 are in good agreement with the recommended values by Matheu et al. and K. Wang et al. At $T = 500\text{--}1500\text{ K}$, our calculated results are less than a factor of 4 with the results of Matheu et al. and a factor of 3 with the results of Wang et al. and at $500\text{--}800\text{ K}$ are a factor of 4.5 with the results of Sirjean et al. Figure 5(c) shows a comparison of the calculated high-pressure limit rate constants for 1,4-H shift reactions $R1 \leftrightarrow R4$ and $R2 \leftrightarrow R5$ in this work with the values of Wang et al. [63], Sirjean et al. [64] and Bankiewicz et al. [65]. It can be seen that our calculated rate constants for 1,4-H shift reactions are in good agreement with the recommended values by Sirjean et al. and Bankiewicz et al.

At $T = 500\text{--}1500\text{ K}$, the calculated results are about a factor of 2.5 lower than the results of Sirjean et al. and Bankiewicz et al. and about a factor of 4.5 higher than the results of Wang et al. Figure 5(d) shows a comparison of the calculated rate constants of β -scission reactions $2\text{-IM-11} \rightarrow \text{C}_2\text{H}_5 + 1,3\text{-C}_4\text{H}_6$ and $6\text{-IM-3} \rightarrow 2\text{-C}_4\text{H}_9 + 1,3\text{-C}_4\text{H}_6$ with the results of Sabbe et al. [66]. At $T = 500\text{--}1500\text{ K}$ and it can be seen that our calculated rate constants are about 4 times lower than the results for reaction $2\text{-IM-11} \rightarrow \text{C}_2\text{H}_5 + 1,3\text{-C}_4\text{H}_6$ and are about a factor of 2 less than the results for reaction $6\text{-IM-3} \rightarrow 2\text{-C}_4\text{H}_9 + 1,3\text{-C}_4\text{H}_6$.

3.2.2. Pressure-dependent rate constants

In the 92 reactions we have calculated high-pressure limit rate constants above, there are six reactions that are

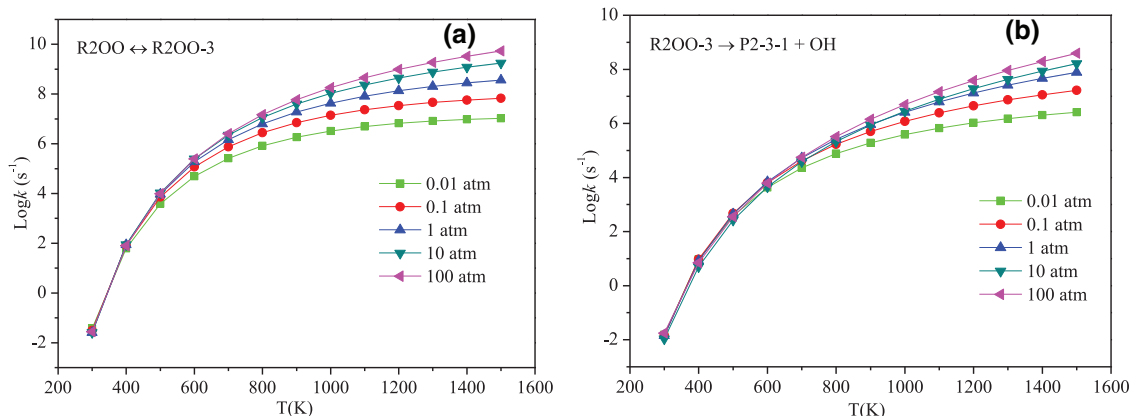


Fig. 6. Plot of the calculated pressure-dependent rate constants of reactions $R200 \leftrightarrow R200-3$ and $R200-3 \rightarrow P2-3-1 + OH$.

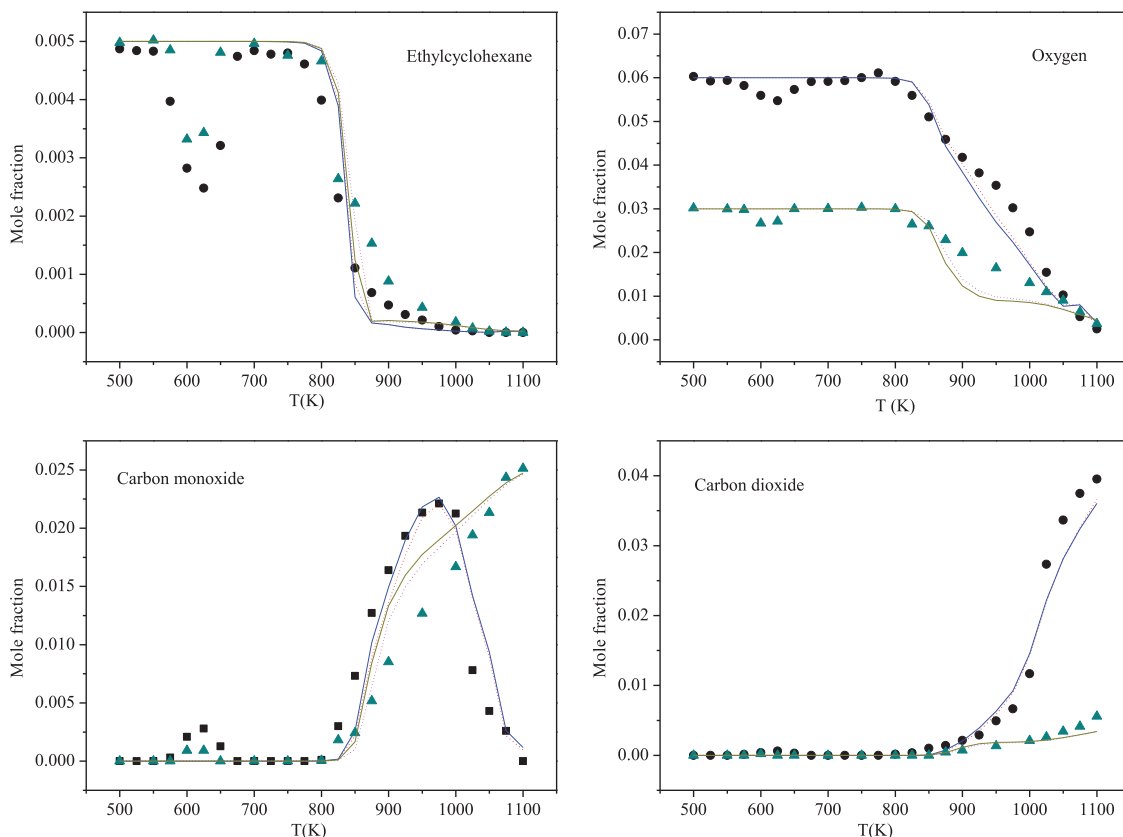


Fig. 7. Comparisons of the concentration profiles for species ECH, O_2 , CO and CO_2 with equivalence ratios: (●) $\phi = 1.0$; (▲) $\phi = 2.0$; our modified mechanism (dot lines), JetSurF 2.0 model [27] (lines) and experiments (symbols) by Husson et al. [24].

pressure-independent and they are the initial H abstraction reactions for the formation of R1–R6. Pressure-dependent rate constants are calculated for the remained 86 pressure-dependent reactions in the temperature range of 300–1500 K and the pressure varying from 0.01 to 100 atm. The rate constants of these pressure-dependent reactions are given in Table S11, S12, S13 and S14 in a modified Arrhenius expression in the Supplemental Material-II. It can be seen from these Tables that the pressure dependence of these reactions are significant. For example, Fig. 6 shows pressure dependence of the rate constants for reactions: $R2O0 \leftrightarrow R2O0-3$ and $R2O0-3 \rightarrow P2-3-1 + OH$. The two reactions correspond to the isomerization of R2O0 and the decomposition of R2O0-3, respectively. The pressure effect becomes significant at temperatures greater than 700 K for the former reaction and at temperatures greater than 1000 K for the latter one. This implies that the pressure effect should not be neglected especially at higher temperatures. Finally, the mechanism of *JetSurF* 2.0 [27] is updated by replacing the rate constants of the reactions in this study with our calculated results and used to simulate the low temperature oxidation of ECH in the jet-stirred reactor experiment ($T = 500$ – 1100 K, $P = 800$ Torr and $\phi = 1.0$ and 2.0) by Husson et al. [24]. The comparison results of the concentration profiles of species ECH, oxygen, carbon monoxide and carbon dioxide are given in Fig. 7. It can be seen that the concentration profiles of ECH, O_2 and CO are slightly improved and the concentration profile of CO_2 our results are close to the results of the *JetSurF* 2.0 model.

4. Conclusions

The low- and intermediate-temperature oxidation mechanisms and kinetics of ECH have been investigated by quantum chemistry calculations in this work. It is shown that in the main products of ECH oxidation, the products having a $C_8H_{14}O$ formula correspond to cyclic ethers and they are mainly 1-oxaspiro[3.5]octane (**P1-2-1**), 8-methyl-7-oxa-bicyclo[4.2.0]octane (**P2-3-1**), 7-methyl-6-oxa-bicyclo[3.2.1]octane (**P2-4-1**) and 2-ethyl-6-oxa-bicyclo[3.1.1]heptane (**P4-5-1**) in low temperature oxidation, and they are different to the experimental results by Husson et al. [20], who have concluded that the main cyclic ethers consist of 7-oxabicyclo[4.3.0]nonane (**P1-3-1**), 3-methyl-2-oxaspiro[1.5]octane (**P2-2-1**), 1-ethyl-7-oxabicyclo[4.1.0]heptane (**P3-3-1**) and 1-methyl-7-oxabicyclo[2.2.1]heptane (**P3-5-1**). For the cycloalkylperoxy radicals (ROO) isomerization to form hydroperoxycycloalkyl radicals (QOOH), it is shown that the 1,5 H-shift is more competitive than the 1,6 H-shift in the cycloalkylperoxy radicals (ROO). In the intermediate temperature, the cycloalkyl radicals (R1–R6) mainly undergo by β -scission and isomerization reactions. The reaction pathways and potential energy surfaces to form the main products (C_2H_4 , 1,3- C_4H_6 , 1- C_4H_8 , methylenecyclopentane (**1-P-3**), 1,3-pentadiene (**2-P-2-2**), methylenecyclohexane (**3-P-1**), 1,3-butadiene-2-ethyl (**3-P-3**), 1,3-hexadiene (**4-P-3**) and so on) of R1–R6 decomposition are also given and the rate constants of relative reactions are calculated by using the TST. It can be seen that the 5-exo ring closure is the most important reaction channel and the main products of ECH oxidation at intermediate temperature are mainly formed by the isomerization and β -scission reactions of the intermediates which are formed by 5-exo ring closure reaction. For the barrierless reaction $R + O_2$, VTST is employed to calculate the rate constant and it can be seen that the calculated rate constants for addition reactions $R1 + O_2$ and $R2 + O_2$, in which the unpaired electron of cyclic C_8H_{15} radicals is in the side-chain, are smaller than $R3 + O_2$, $R4 + O_2$, $R5 + O_2$ and $R6 + O_2$, in which the unpaired electron of cyclic C_8H_{15} radicals is in the ring. Finally, rate constants of pressure-dependent reactions in the three-parameter modified Arrhenius expression are derived in the temperature range of 300–1500 K. In addition, the kinetics at low- and intermediate-temperature of ECH oxidation in this work can be

helpful to improve the popular *JetSurF* 2.0 and the chemical kinetic modeling of the low temperature oxidation of cycloalkanes.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2015.08.004.

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